Review plan · for the lecturer

Teaching networks and sequences: unit check

University

Time
40 min

Materials
Paper; The four lessons

Learning goals

- A forward step and a descent step by hand
- The smallest design for each job
- What the Eight notes lab did not prove

Activity: Work it

1. Alone (15 min): Answer the ten questions with no notes, and write one line on why for each.
2. By hand (10 min): Redo the forward step and the descent step on paper, then compare your working with the explanations.
3. Read (10 min): Read every explanation, the ones you got right too, and mark any reason that differs from yours.
4. Plan (5 min): Name the idea you will redo and the lesson it comes from.

Discussion

- Which lesson was the miss?

Exit ticket

- Name the idea you will redo.

Try it in your next lecture
Redo the one item you missed, on paper, before the next course.

Quiz sheet · for students

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Name | Group | Date

1 **Inputs x = (2, 1). A hidden ReLU neuron has weights (1, 0.5) and bias −1. What does it output?**

◯ 0 ◯ 1.5 ◯ 2.5

2 **Two linear layers with no activation between them can draw only one straight cut, just like a single layer.**

◯ True ◯ False

3 **Match the job to the smallest design that does it.**

Split two groups with one straight line	• • • •	A hidden layer with an activation
Separate points inside a ring from those outside	• • • •	Attention
Carry a short pattern along a sequence, one symbol at a time	• • • •	A single neuron
Use a word 30 positions back without it fading	• • • •	A recurrent net

4 **Loss θ^2, $\theta = 5$, $\eta = 0.1$. θ after one step?**

◯ 4 ◯ 4.5 ◯ 6

5 **If the training loss keeps falling, the model will be right on new data.**

◯ True ◯ False

6 **Loss θ^2, starting at $\theta = 1$. Match the learning rate to where one step lands.**

$\eta = 0.1$	• • • •	θ lands twice as far out, on the other side
$\eta = 0.5$	• • • •	θ moves a little closer, to 0.8
$\eta = 1.5$	• • • •	θ lands on the bottom, 0

7 **In the lab "A memory that fades", which of these is the smallest memory that still leaves more than half the cue at the end?**

◯ 0.75 ◯ 0.8 ◯ 0.9

8 **A longer gap between the cue and the end needs a memory closer to 1 to keep the same share of the cue.**

◯ True ◯ False

9 **Match what it does to the part that does it.**

Reads one symbol at a time and carries a state	• • • •	The residual connection
Scores every earlier item and mixes them	• • • •	A recurrent net
Adds the original back after the mix	• • • •	Attention

10 **After the tune C E A E C E A E, the Eight notes lab writes A E A E. Why does C never come back?**

◯ After E it counted A twice and C once, and it always takes the top count

◯ It ran out of training steps before it could reach C ◯ It learned from the tune that C sounds wrong after E

Answers · for the lecturer

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1 **1.5**

1 × 2 + 0.5 × 1 − 1 = 1.5, and ReLU keeps a positive sum as it is.

2 **True**

With no activation the two layers multiply out to one linear layer: one straight cut, and no straight cut separates a ring.

3 **Split two groups with one straight line — A single neuron**
Separate points inside a ring from those outside — A hidden layer with an activation
Carry a short pattern along a sequence, one symbol at a time — A recurrent net
Use a word 30 positions back without it fading — Attention

One neuron makes one cut, a ring needs the bend of an activation, a short sequence needs only a carried state, and a link 30 positions back needs attention's direct look back.

4 **4**

The slope is 2θ = 10, so the step is 0.1 × 10 = 1 and θ becomes 4.

5 **False**

A falling training loss can be memorising.

6 **η = 0.1 — θ moves a little closer, to 0.8**
η = 0.5 — θ lands on the bottom, 0
η = 1.5 — θ lands twice as far out, on the other side

One step multiplies θ by 1 − 2η: 0.8 for η = 0.1, 0 for η = 0.5, and −2 for η = 1.5, twice as far out on the other side.

7 **0.8**

The cue is multiplied by the memory at each of the three empty steps: 0.8 leaves 0.512, just over half, and 0.75 leaves about 0.42.

8 **True**

The share left is the memory multiplied by itself once per empty step, so more steps need a memory nearer 1.

9 **Reads one symbol at a time and carries a state — A recurrent net**
Scores every earlier item and mixes them — Attention
Adds the original back after the mix — The residual connection

The recurrent net hopes its state kept the start, attention looks back, and the residual connection keeps what was there.

10 **After E it counted A twice and C once, and it always takes the top count**

It only counted which note followed which, and it always takes the top count.